TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	73.3 ± 6.13
Correlation Stability	85.7 ± 2.05
Distribution Consistency	85.3 ± 2.49
Entropy Preservation	94.7 ± 0.47
Mean Z Score Drift	0.2 ± 0.03
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	27.3 ± 7.41
Data Density	100.0 ± 0.00
Coverage Score	58.0 ± 14.51
Feature Accuracy	88.7 ± 1.25
Novelty Score	0.5 ± 0.09

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Dataset Audit: parkinsons.csv

1. Dataset Portrait

Attribute	Value
Rows	195
Columns	23
Numeric features	23
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4.1
Folds (*13)	3
Context % (*14)	20%
Generation % (*15)	100%
Join context (*16)	True

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
MDVP:Fo	154.229	138.961	-15.267	9.9%	OK
MDVP:Fhi	197.105	163.626	-33.479	17.0%	Warn
MDVP:Flo	116.325	99.487	-16.838	14.5%	Warn
MDVP:Jitter	0.006	0.007	+0.001	15.4%	Warn
MDVP:Jitter.1	0.006	0.007	+0.001	15.3%	Warn
MDVP:RAP	0.003	0.004	+0.001	17.4%	Warn
MDVP:PPQ	0.003	0.004	+0.001	16.8%	Warn
Jitter:DDP	0.010	0.012	+0.002	17.4%	Warn
MDVP:Shimmer	0.030	0.035	+0.005	17.5%	Warn
MDVP:Shimmer.1	0.030	0.035	+0.005	17.6%	Warn
Shimmer:APQ3	0.016	0.018	+0.002	15.8%	Warn
Shimmer:APQ5	0.018	0.020	+0.003	14.2%	Warn

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	73.3 ± 6.13
Correlation Stability (*2)	85.7 ± 2.05
Distribution Consistency (*3)	85.3 ± 2.49
Entropy Preservation (*4)	94.7 ± 0.47
Mean Z Score Drift (*5)	0.2 ± 0.03
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	27.3 ± 7.41

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	58.0 ± 14.51
Feature Accuracy (*10)	88.7 ± 1.25
Novelty Score (*11)	0.5 ± 0.09

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9743589743589743	0.8461538461538461	0.9829059829059833	0.008547008547008553	
F1	0.9806435394670692	0.8813923673612102	0.9866613311991462	0.006017791732077406	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8888888888888889	0.8376068376068376	0.8205128205128206	-0.0683760683760684	
F1	0.9167514124293786	0.8707154359328272	0.8638262322472848	-0.05292518018209369	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9487179487179486	0.8034188034188036	0.9487179487179488	0	
F1	0.9622806387512269	0.8411802705578354	0.963702147525677	0.001421508774449946	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9572649572649571	0.7692307692307692	0.8888888888888889	-0.06837606837606831	
F1	0.968042328042328	0.818042328042328	0.9169132133727808	-0.05112911466954726	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7777777777777778	0.7008547008547009	0.7777777777777778	0	
F1	0.8614267676767677	0.8213028169014085	0.8614267676767677	0	

Model: LR

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8803418803418803	0.7692307692307692	0.8803418803418803	0	
F1	0.91232423490488	0.8388278388278388	0.9092473118279569	-0.00307692307692308	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8461538461538463	0.6153846153846154	0.8376068376068376	-0.00854700854700859 1	
F1	0.891808924827792 7	0.7056443191311613	0.881410934744268	-0.01039799008352471	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.717948717948718	0.7692307692307692	0.6923076923076924	-0.02564102564102566	
F1	0.749906191369606	0.816933638443936	0.7266944941363546	-0.02321169723325141	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8888888888888888	0.6923076923076924	0.8717948717948718	-0.01709401709401707	
F1	0.9177859177859178	0.7918356037866543	0.9020346600178533	-0.01575125776806448	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9401709401709403	0.8376068376068376	0.9487179487179486	0.00854700854700851 7	
F1	0.9570703384745131	0.8707879508816277	0.9625806451612903	0.00551030668677724 8	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	119.992	157.302	74.997	0.00784	0.00784	0.0037	0.00554	0.01109	0.04374	0.04374
2	122.4	148.65	113.819	0.00968	0.00968	0.00465	0.00696	0.01394	0.06134	0.06134
3	116.682	131.111	111.555	0.0105	0.0105	0.00544	0.00781	0.01633	0.05233	0.05233
4	116.676	137.871	111.366	0.00997	0.00997	0.00502	0.00698	0.01505	0.05492	0.05492
5	116.014	141.781	110.655	0.01284	0.01284	0.00655	0.00908	0.01966	0.06425	0.06425
6	120.552	131.162	113.787	0.00968	0.00968	0.00463	0.0075	0.01388	0.04701	0.04701
7	120.267	137.244	114.82	0.00333	0.00333	0.00155	0.00202	0.00466	0.01608	0.01608
8	107.332	113.84	104.315	0.0029	0.0029	0.00144	0.00182	0.00431	0.01567	0.01567
9	95.73	132.068	91.754	0.00551	0.00551	0.00293	0.00332	0.0088	0.02093	0.02093
10	95.056	120.103	91.226	0.00532	0.00532	0.00268	0.00332	0.00803	0.02838	0.02838
11	88.333	112.24	84.072	0.00505	0.00505	0.00254	0.0033	0.00763	0.02143	0.02143
12	91.904	115.871	86.292	0.0054	0.0054	0.00281	0.00336	0.00844	0.02752	0.02752
13	136.926	159.866	131.276	0.00293	0.00293	0.00118	0.00153	0.00355	0.01259	0.01259
14	139.173	179.139	76.556	0.0039	0.0039	0.00165	0.00208	0.00496	0.01642	0.01642
15	152.845	163.305	75.836	0.00294	0.00294	0.00121	0.00149	0.00364	0.01828	0.01828
16	142.167	217.455	83.159	0.00369	0.00369	0.00157	0.00203	0.00471	0.01503	0.01503
17	144.188	349.259	82.764	0.00544	0.00544	0.00211	0.00292	0.00632	0.02047	0.02047
18	168.778	232.181	75.603	0.00718	0.00718	0.00284	0.00387	0.00853	0.03327	0.03327
19	153.046	175.829	68.623	0.00742	0.00742	0.00364	0.00432	0.01092	0.05517	0.05517
20	156.405	189.398	142.822	0.00768	0.00768	0.00372	0.00399	0.01116	0.03995	0.03995

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B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	193.03	208.9	80.297	0.00766	0.00766	0.0045	0.00389	0.01351	0.03044	0.03044
2	115.38	123.109	108.634	0.00332	0.00332	0.0016	0.00199	0.0048	0.01503	0.01503
3	154.003	160.267	128.621	0.00183	0.00183	0.00076	0.001	0.00229	0.0103	0.0103
4	95.056	120.103	91.226	0.00532	0.00532	0.00268	0.00332	0.00803	0.02838	0.02838
5	214.289	260.277	77.973	0.00567	0.00567	0.00295	0.00317	0.00885	0.01884	0.01884
6	237.323	243.709	229.256	0.00303	0.00303	0.00173	0.00159	0.00519	0.01242	0.01242
7	142.37	180.52	92.13	0.00611	0.00611	0.00307	0.00332	0.00921	0.02547	0.02547
8	125.49	136.74	112.23	0.00378	0.00378	0.00185	0.00213	0.00554	0.01713	0.01713
9	158.62	169.18	140.41	0.00421	0.00421	0.00213	0.00234	0.00639	0.02876	0.02876
10	132.97	149.61	90.74	0.00714	0.00714	0.00354	0.00402	0.01061	0.03092	0.03092
11	185.24	220.13	80.12	0.00843	0.00843	0.00421	0.00498	0.01263	0.02297	0.02297
12	111.56	120.87	106.12	0.00289	0.00289	0.00121	0.00141	0.00362	0.01209	0.01209
13	110.417	131.067	103.37	0.00784	0.00784	0.00352	0.00514	0.01056	0.03715	0.03715
14	208.083	253.792	91.802	0.00757	0.00757	0.00428	0.00428	0.01285	0.06725	0.06725
15	244.99	272.21	239.17	0.00451	0.00451	0.00279	0.00237	0.00837	0.01897	0.01897
16	228.969	239.541	113.201	0.00238	0.00238	0.00136	0.0014	0.00408	0.01745	0.01745

B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	120.08	139.71	111.208	0.00405	0.00405	0.0018	0.0022	0.0054	0.01706	0.01706
2	186.695	211.961	149.442	0.01872	0.01872	0.01075	0.01154	0.03225	0.0583	0.0583
3	197.076	206.896	192.055	0.00289	0.00289	0.00166	0.00168	0.00498	0.01098	0.01098
4	223.365	238.987	98.664	0.00264	0.00264	0.00154	0.00151	0.00461	0.01906	0.01906
5	169.774	191.759	151.451	0.01568	0.01568	0.00863	0.00946	0.02589	0.08143	0.08143
6	95.385	102.145	90.264	0.00608	0.00608	0.00331	0.00332	0.00994	0.03202	0.03202
7	137.56	151.43	120.11	0.00521	0.00521	0.00209	0.00265	0.00627	0.02438	0.02438
8	190.72	215.67	162.34	0.01389	0.01389	0.00723	0.00815	0.02169	0.06284	0.06284
9	124.31	132.85	116.42	0.00409	0.00409	0.00136	0.00156	0.00408	0.01593	0.01593
10	102.68	110.43	95.12	0.00672	0.00672	0.00352	0.00391	0.01056	0.02841	0.02841
11	210.33	228.44	185.22	0.00317	0.00317	0.00121	0.00134	0.00363	0.01852	0.01852
12	117.95	141.12	109.45	0.00912	0.00912	0.00429	0.00511	0.01287	0.04317	0.04317
13	245.51	262.09	231.848	0.00235	0.00235	0.00127	0.00148	0.0038	0.01608	0.01608
14	115.38	123.109	108.634	0.00332	0.00332	0.0016	0.00199	0.0048	0.01503	0.01503
15	95.056	120.103	91.226	0.00532	0.00532	0.00268	0.00332	0.00803	0.02838	0.02838
16	143.533	162.215	65.809	0.01101	0.01101	0.00647	0.00467	0.01941	0.05384	0.05384

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B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	170.368	268.796	79.543	0.00571	0.00571	0.00232	0.00269	0.00696	0.03273	0.03273
2	102.273	142.83	85.902	0.00907	0.00907	0.00493	0.00461	0.0148	0.02814	0.02814
3	120.552	131.162	113.787	0.00968	0.00968	0.00463	0.0075	0.01388	0.04701	0.04701
4	122.336	142.369	94.794	0.00727	0.00727	0.00415	0.00339	0.01246	0.02293	0.02293
5	100.96	110.019	95.628	0.00606	0.00606	0.00351	0.00348	0.01053	0.02427	0.02427
6	109.86	126.358	104.437	0.00874	0.00874	0.00398	0.00539	0.01193	0.03209	0.03209
7	134.217	155.982	90.326	0.00612	0.00612	0.00318	0.00341	0.00954	0.02567	0.02567
8	112.439	132.184	100.285	0.00437	0.00437	0.00221	0.00247	0.00662	0.01988	0.01988
9	157.893	170.416	78.539	0.00789	0.00789	0.00432	0.00466	0.01295	0.03312	0.03312
10	124.008	143.115	93.477	0.00597	0.00597	0.00293	0.00312	0.00878	0.02143	0.02143
11	141.563	161.784	70.128	0.01134	0.01134	0.00629	0.00601	0.01887	0.04712	0.04712
12	119.845	129.473	112.019	0.00321	0.00321	0.00135	0.00173	0.00406	0.01387	0.01387
13	242.852	255.034	227.911	0.00225	0.00225	0.00117	0.00139	0.0035	0.01494	0.01494
14	139.224	586.567	66.157	0.03011	0.03011	0.01854	0.01628	0.05563	0.09419	0.09419
15	186.163	197.724	177.584	0.00298	0.00298	0.00165	0.00175	0.00496	0.01495	0.01495
16	116.879	131.897	108.153	0.00788	0.00788	0.00334	0.00493	0.01003	0.02645	0.02645